

Athanasios(Thanos) Papaioannou

Senior Research Scientist & Generative AI Engineer

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RESEARCH PROFILE

Senior computer vision researcher with 10+ years of experience building large-scale 3D and 4D visual systems, neural rendering pipelines, and generative models for human-centric applications. Track record of first- and senior-author publications at top venues (ECCV, ICCV, IJCV, *Scientific Reports*), with industry experience translating research into production-grade systems. Current work centres on neural rendering, Gaussian splatting, and feed-forward 3D reconstruction. Broader research interests extend toward dynamic 4D scene modelling, world models, and agentic perception – connecting predictive 3D representations with embodied reasoning and interaction.

TECHNICAL EXPERTISE

3D / 4D Vision Neural rendering, Gaussian splatting (3DGS, feed-forward variants), neural fields, differentiable rendering, novel-view synthesis, 3D/4D reconstruction, statistical and morphable shape models, mesh-based representations.

Generative & Representation Learning Autoencoders, self-supervised learning, multimodal representation learning, foundation models for vision.

Research Interests World models, vision-language and vision-language-action (VLA) systems, agentic perception, dynamic scene understanding, spatial reasoning, embodied AI.

ML Systems & Methodology Large-scale dataset construction, distributed and multi-GPU training, evaluation and ablation design, experimentation frameworks, research-to-production workflows.

Programming Python, C++ (production); previously MATLAB.

Frameworks & Tools PyTorch (primary), NumPy, scikit-learn, Git, Linux, Slurm, GPU/cluster computing.

EDUCATION

PhD, Computing — Imperial College London	2012 – 2016
MSc, Informatics — Aristotle University of Thessaloniki	2003 – 2005
BSc, Informatics — Aristotle University of Thessaloniki	1999 – 2003

PROFESSIONAL EXPERIENCE

Research Software Engineer, Generative AI — Google, London Sep 2026 – Present

Senior Research Engineer — Huawei Research, London Jun 2020 – Sep 2026

- Lead computer vision research on neural rendering, 3D/4D reconstruction, and generative modelling of humans and scenes, including *feed-forward Gaussian splatting* for novel-view synthesis.
- Designed and trained large-scale generative 3D models (statistical, mesh-based, and implicit morphable representations) for high-fidelity human modelling.

- Built end-to-end data pipelines for multi-view, RGB-D, and 4D capture data, supporting model training, evaluation, and deployment on complex visual datasets.
- Drove evaluation and ablation methodology for reconstruction quality, geometric accuracy, and view-consistency, enabling systematic model comparison and iteration.
- Translated research prototypes into integration-ready systems in collaboration with infrastructure and engineering teams; supervised junior researchers and contributed to research direction setting.
- Co-organised the View Synthesis Challenge for Human Heads at ICCVW 2023.

Senior Research Engineer — Facesoft, London

Nov 2019 – Jun 2020

- Developed deep learning models for face analysis, recognition, and representation learning, deployed in real-time production computer vision systems.
- Designed data processing and training pipelines for large-scale face datasets, with emphasis on robustness across demographics and capture conditions.

Research Associate — Institute of Child Health, UCL, London

Jan 2018 – Oct 2019

- Developed statistical and deep learning methods for analysing complex 3D craniofacial datasets, including mesh autoencoders for clinical 3D shape analysis.
- Built data processing and evaluation workflows for clinical 3D datasets and collaborated with surgeons, clinicians, and researchers on translational ML projects.
- Contributed to peer-reviewed publications on 3D shape modelling and clinical decision support.

Research Associate — Imperial College London

Jan 2017 – Dec 2018

- Developed face tracking and verification systems for secure online assessment platforms, including identity verification and behavioural analysis components deployed at scale.

Research Assistant — Imperial College London

Oct 2012 – Dec 2016

- Built large-scale 4D facial datasets for emotion and behaviour recognition research; developed algorithms for automated facial analysis and statistical 3D modelling.
- Contributed to national and European research projects on face and behaviour analysis.

SELECTED PUBLICATIONS

- 2026** R. Shaw*, Y. Jang*, A. Papaioannou*, A. Moreau*, H. Dharmo, Z. Zhang, E. Perez-Pellitero. *ICo3D: An Interactive Conversational 3D Virtual Human*. IJCV, 134, 161, 2026. (*joint first authorship)
- 2025** R. Potamias, S. Galanakis, J. Deng, A. Papaioannou, S. Zafeiriou. *imHead: A Large-Scale Implicit Morphable Model for Localized Head Modeling*. ICCV 2025.
- 2022** A. Papaioannou, B. Gecer, S. Cheng, et al. *MimicME: A Large Scale Diverse 4D Database for Facial Expression Analysis*. ECCV 2022.
- 2022** E. O'Sullivan, L. S. van de Lande, A. Papaioannou, et al. *Convolutional Mesh Autoencoders for the 3-Dimensional Identification of FGFR-related Craniosynostosis*. Scientific Reports, 12, 2230, 2022.
- 2020** B. Gecer, A. Lattas, S. Ploumpis, J. Deng, A. Papaioannou, S. Moschoglou, S. Zafeiriou. *Synthesizing Coupled 3D Face Modalities by Trunk-Branch Generative Adversarial Networks*. ECCV 2020.

Full publication list: [Google Scholar profile](#).

RESEARCH LEADERSHIP & PROFESSIONAL SERVICE

- **Challenge Organiser** – View Synthesis Challenge for Human Heads, ICCVW 2023.
- **Local Chair & Web Chair** – British Machine Vision Conference (BMVC) 2017.
- **Reviewer** – CVPR, ECCV, ICCV, ACM Multimedia, BMVC, Scientific Reports.
- **Research supervision** – mentoring of junior researchers and engineers within applied CV research projects.